

SQL Statements

PAGE 1 – Introduction to SQL

SQL – Structured Query Language

Definition: SQL (Structured Query Language) is a standard programming language used to create, manage, manipulate, and retrieve data from relational databases.

SQL is widely used with database management systems such as MySQL, Oracle, PostgreSQL, and SQL Server.

Uses of SQL

1. Creating databases and tables
2. Inserting data into tables
3. Updating existing data
4. Deleting unwanted data
5. Retrieving required information
6. Managing user permissions
7. Managing database transactions

Types of SQL Statements

SQL statements are commonly classified into five categories: DDL, DML, DQL, DCL, and TCL. Each category is used for a different type of database operation.

PAGE 2 – DDL

DDL – Data Definition Language

Definition: DDL is used to define, create, and modify the structure of database objects such as tables.

1. CREATE

CREATE is used to create a new database object such as a table.

```
CREATE TABLE student (  
    id INT,  
    name VARCHAR(50),  
    age INT  
);
```

2. ALTER

ALTER is used to modify the structure of an existing table.

```
ALTER TABLE student ADD email VARCHAR(100);
```

3. DROP

DROP is used to permanently remove a database object.

```
DROP TABLE student;
```

4. TRUNCATE

TRUNCATE removes all records from a table while keeping the table structure.

```
TRUNCATE TABLE student;
```

DDL Summary: CREATE – creates a table; ALTER – modifies structure; DROP – deletes table/object; TRUNCATE – removes all table data.

PAGE 3 – DML & DQL

DML – Data Manipulation Language

Definition: DML is used to add, modify, and remove data stored in database tables.

1. INSERT

INSERT is used to add new records.

```
INSERT INTO student (id, name, age)
VALUES (1, 'Subeka', 20);
```

2. UPDATE

UPDATE is used to modify existing records.

```
UPDATE student SET age = 21 WHERE id = 1;
```

3. DELETE

DELETE is used to remove records from a table.

```
DELETE FROM student WHERE id = 1;
```

DQL – Data Query Language

Definition: DQL is used to retrieve or fetch data from a database.

SELECT is the main DQL command.

```
SELECT * FROM student;
```

```
To retrieve selected columns: SELECT name, age FROM student;
```

PAGE 4 – DCL

DCL – Data Control Language

Definition: DCL is used to control access and permissions of users in a database.

DCL is mainly used by database administrators to provide or remove privileges.

1. GRANT

GRANT is used to give specific permissions to a user.

```
GRANT SELECT ON student TO user1;
```

This gives user1 permission to retrieve data from the student table.

Another example:

```
GRANT INSERT ON student TO user1;
```

2. REVOKE

REVOKE is used to remove previously given permissions.

```
REVOKE SELECT ON student FROM user1;
```

This removes the SELECT permission from user1.

Importance of DCL

- Protects database information
- Controls user access
- Prevents unauthorized operations
- Improves database security
- Assigns different privileges to different users

PAGE 5 – TCL & Overall Summary

TCL – Transaction Control Language

Definition: TCL is used to manage transactions and control changes made to a database.

A transaction is a group of SQL operations treated as a single unit.

1. COMMIT

COMMIT permanently saves the changes made during a transaction.

```
INSERT INTO student VALUES (2, 'Arun', 21);  
COMMIT;
```

2. ROLLBACK

ROLLBACK is used to undo changes that have not been committed.

```
UPDATE student SET age = 22 WHERE id = 2;  
ROLLBACK;
```

3. SAVEPOINT

SAVEPOINT creates a temporary point inside a transaction.

```
SAVEPOINT sp1;  
ROLLBACK TO sp1;
```

Overall SQL Statement Summary

Category	Full Form	Commands
DDL	Data Definition Language	CREATE, ALTER, DROP, TRUNCATE
DML	Data Manipulation Language	INSERT, UPDATE, DELETE
DQL	Data Query Language	SELECT
DCL	Data Control Language	GRANT, REVOKE
TCL	Transaction Control Language	COMMIT, ROLLBACK, SAVEPOINT

Easy Memory: DDL → Structure | DML → Data Change | DQL → Data Retrieve | DCL → Permission |
TCL → Transaction